

Hotel Booking Customer Retention and Dynamic Pricing Analysis

Week 4 Report

Business Insights and Recommendations

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Tools Used:

- Python
 - SQL
 - Power BI
 - Git & GitHub
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1. Introduction

The fourth phase of the project focused on interpreting the Power BI dashboard to generate actionable business insights and strategic recommendations. Using the visualizations and key performance indicators developed during the dashboard phase, this report identifies booking patterns, customer behavior, cancellation trends, and pricing opportunities. The objective is to support hotel management in making informed decisions that improve customer retention and optimize revenue.

2. Dashboard Summary

The Power BI dashboard provides an interactive overview of hotel booking performance through key performance indicators, charts, and slicers. It enables users to monitor booking trends, compare hotel types, analyze cancellations, and evaluate pricing metrics. The dashboard was designed to simplify decision-making by presenting important business information in a clear and interactive format.

The dashboard includes:

- Five KPI Cards
 - Monthly Booking Trend
 - Hotel Type Comparison
 - Cancellation Distribution
 - Interactive Slicers
 - Business Insights Section
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3. Key Business Insights

Based on the dashboard analysis, the following insights were identified:

Seasonal Booking Trends

Hotel bookings increase significantly during the mid-year months, with July and August recording the highest booking volumes. This indicates a strong seasonal demand pattern and highlights periods where hotels should prepare for increased occupancy.

Hotel Type Performance

City Hotels receive a higher number of bookings compared to Resort Hotels. However, they also experience a higher number of booking cancellations, indicating the need for targeted customer retention strategies.

Cancellation Patterns

Customers with longer lead times are more likely to cancel their bookings. Understanding this behavior can help hotels implement flexible booking policies and timely customer engagement to reduce cancellations.

Pricing Analysis

Average Daily Rate (ADR) increases during periods of high demand, reflecting the effectiveness of demand-based pricing strategies. Monitoring ADR trends allows hotels to optimize room pricing while remaining competitive.

Customer Behavior

Customer booking patterns reveal opportunities to improve guest retention through personalized offers, loyalty programs, and targeted promotions.

4. Business Recommendations

Based on the insights generated from the dashboard, the following recommendations are proposed:

- Increase staffing and room availability during peak booking months.
 - Implement targeted retention campaigns for customers with longer lead times.
 - Apply dynamic pricing strategies to maximize revenue during high-demand periods.
 - Introduce promotional offers during low-demand seasons to improve occupancy.
 - Monitor cancellation trends regularly and optimize booking policies.
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5. Business Impact

The Power BI dashboard provides decision-makers with a centralized platform for monitoring hotel performance. By combining operational metrics with visual analytics, management can identify trends quickly, improve pricing strategies, reduce cancellations, and enhance customer satisfaction.

The interactive nature of the dashboard enables users to explore data dynamically, making it easier to respond to changing business conditions and support strategic planning.

6. Future Enhancements

Future improvements to the project may include:

- Customer segmentation using machine learning techniques.
 - Booking demand forecasting.
 - Revenue prediction models.
 - Real-time dashboard integration.
 - Automated reporting and scheduled dashboard refresh.
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7. Conclusion

The Hotel Booking Customer Retention and Dynamic Pricing Analysis project successfully demonstrates the application of data analytics, business intelligence, and visualization techniques to solve real-world business problems. Through Python for data preparation, SQL for analysis, and Power BI for dashboard development, the project delivers meaningful insights into booking behavior, pricing trends, and cancellation patterns.

The dashboard enables hotel management to make informed decisions that improve operational efficiency, strengthen customer retention, and optimize revenue. The project also highlights the value of combining technical analysis with business-focused reporting to support strategic decision-making.